

STA 2101: Statistics & Probability

Lecture 03: Levels of Measurement, Frequency

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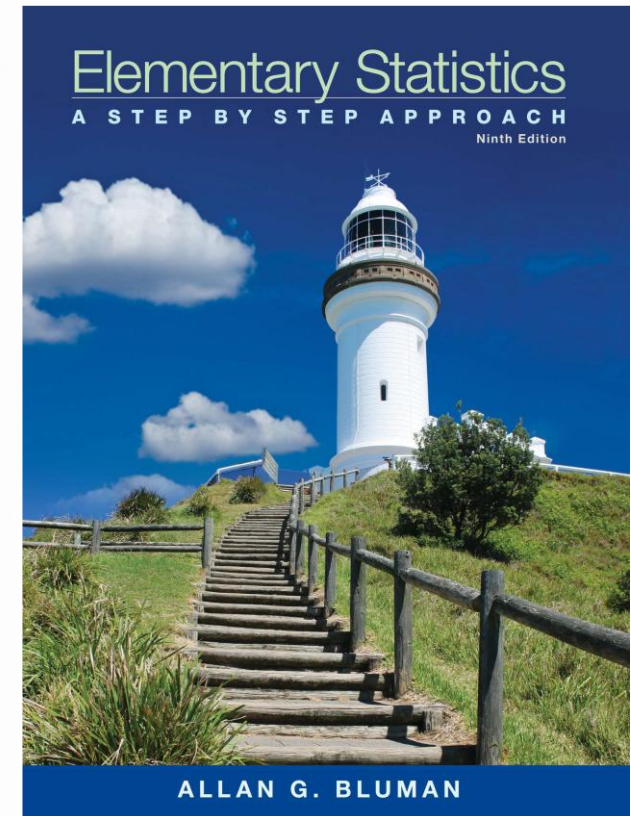
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Class Code
Sec 1 - es2uztok
Sec 5 - ayvzsvgm

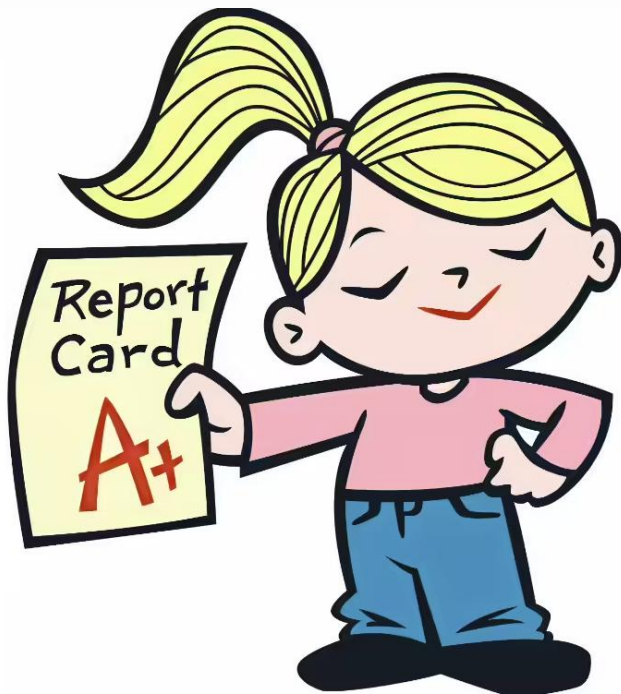
Spring 2026

Reference Book

- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution



Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

Today's Topics

- Levels of Measurement
- Frequency

Before we move... remember

- Time
- What kind of variable is it ???



Before we move... remember

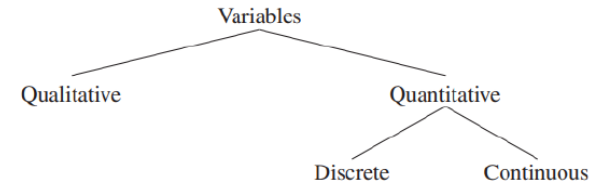
- Time
- Quantitative
- Continuous



Levels of Measurement

- In addition to being classified as qualitative or quantitative, variables can be classified by how they are categorized, counted, or measured.

The classification of variables can be summarized as follows:

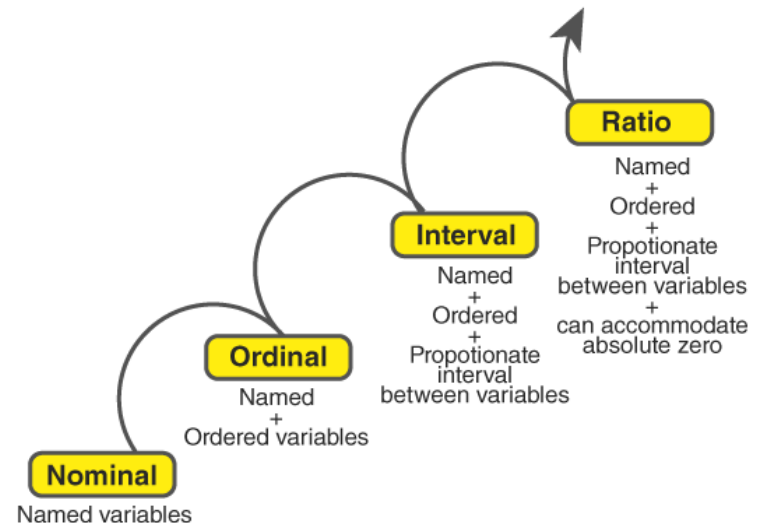


Nominal level of measurement:

This is the lowest level of measurement, where data are categorized without any order or ranking. **The categories are distinct and mutually exclusive.**

Example: Classifying survey subjects as male or female is another example of nominal-level measurement. Classifying residents according to zip codes is also an example of the nominal level of measurement.

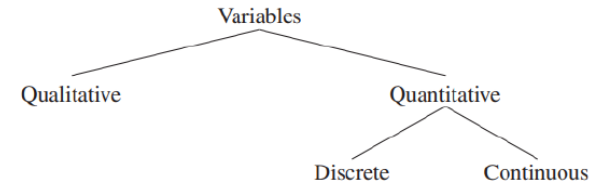
LEVELS OF MEASUREMENT



Levels of Measurement

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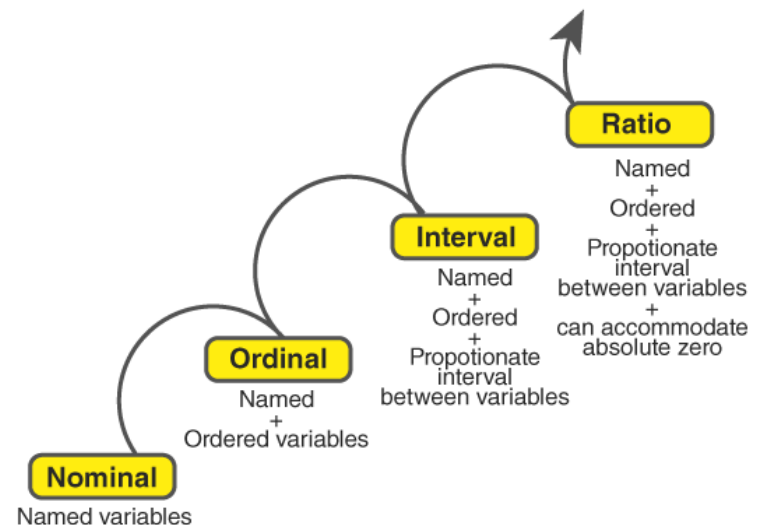
"Distinct" means separate, clearly different from each other.

For example, if categories are "Dog" and "Cat," they don't overlap—they are unique groups.

"Mutually exclusive" means one thing can belong to only one category at a time.

Example: A person cannot be both "male" and "female" in the same survey response. Similarly, a house has only one zip code, not two at once.

LEVELS OF MEASUREMENT



Levels of Measurement (Contd.)

- **Ordinal level of measurement:** The ordinal level of measurement **classifies data into categories that can be ranked**; however, **precise differences between the ranks do not exist**
- Example:
- Rankings in a competition: 1st place, 2nd place, 3rd place, etc. Military Rank: Private, Sergeant, Lieutenant, Captain Educational Level: High school, Bachelor's, Master's, Doctorate Other examples of ordinal data are letter grades (A, B, C, D, F)

Levels of Measurement

- **Interval Level of Measurement**

- **Definition:**

The interval level ranks data, and precise differences between units of measure exist.

However, there is **no meaningful zero**.

- **Examples:**

- **Temperature (Celsius / Fahrenheit)**

- **Ordered:** $10^{\circ}\text{C} < 20^{\circ}\text{C} < 30^{\circ}\text{C}$
 - **Equal Differences:** The difference between 10°C and 20°C is the same as between 20°C and 30°C (both are 10 degrees).
 - **No True Zero:** 0°C or 0°F does not mean "no temperature"; it is just another point on the scale.

- **IQ Scores**

- **Ordered:** $80 < 100 < 120$
 - **Equal Differences:** The difference between 100 and 120 is the same as between 80 and 100 (both are 20 points).
 - **No True Zero:** An IQ score of 0 does not mean "no intelligence"; it is simply another point on the scale.

Levels of Measurement

- **Ratio Level of Measurement**

- **Definition:**

The ratio level is the **highest level of measurement** in statistics.

It has all the properties of interval data (ordered, equal differences) **plus a true zero point.**

- **Examples:**

- **Height**

- **Ordered:** $150\text{ cm} < 160\text{ cm} < 170\text{ cm}$
 - **Equal Differences:** The difference between 160 cm and 170 cm is the same as between 150 cm and 160 cm (both are 10 cm).
 - **True Zero:** A height of 0 cm represents the complete absence of height.
 - **Ratios are Meaningful:** A person who is 180 cm tall is twice as tall as someone who is 90 cm tall.

Levels of Measurement

- **Ratio Level of Measurement**

- **Definition:**

The ratio level is the **highest level of measurement** in statistics.

It has all the properties of interval data (ordered, equal differences) **plus a true zero point.**

- **Examples:**

- **Age**

- **Ordered:** 10 years < 20 years < 30 years
 - **Equal Differences:** The difference between 20 years and 30 years is the same as between 10 years and 20 years (both are 10 years).
 - **True Zero:** Age can be 0 years at the exact moment of birth.
 - **Ratios are Meaningful:** Someone who is 40 years old is twice as old as someone who is 20 years old.

Practice Time – Which Level of Measurement do they fall into ?

- **Examples:**

- Blood group (A, B, AB, O)
- Ranking of athletes in a race (1st, 2nd, 3rd)
- Calendar years (1990, 2000, 2010)
- Students' marks in an exam (out of 100)
- Phone numbers
- Movie ratings (★ 1 star, ★ ★ 2 stars, ★ ★ ★ 3 stars, etc.)
- Monthly salary in dollars
- Time of day (2:00 pm, 3:00 pm, 4:00 pm)
- Marital status (single, married, divorced, widowed)
- Temperature in Celsius

- Blood group → **Nominal**
- Ranking of athletes → **Ordinal**
- Calendar years → **Interval**
- Students' marks → **Ratio**
- Phone numbers → **Nominal**
- Movie ratings → **Ordinal**
- Monthly salary → **Ratio**
- Time of day → **Interval**
- Marital status → **Nominal**
- Temperature (Celsius) → **Interval**

Answers

•Blood group (A, B, AB, O) → Nominal

- Just categories (no order).
- $A \neq B$, but one is not "greater" or "lesser" than the other.

•Ranking of athletes (1st, 2nd, 3rd) → Ordinal

- There is an order.
- But we don't know the actual difference (e.g., how many seconds apart).

•Calendar years (1990, 2000, 2010) → Interval

- Ordered, equal differences (2000–2010 = 10 years).
- But "year 0" does not mean "no time."

•Students' marks in an exam → Ratio

- Ordered, equal differences, and **true zero** (0 marks = no score).
- Ratios are meaningful (80 marks is double 40 marks).

•Phone numbers → Nominal

- Numbers are just labels for identification, no mathematical meaning.

•Movie ratings (1 star, 2 stars, 3 stars, etc.) → Ordinal

- Ordered, but the difference is not always equal (difference between 1★ & 2★ may not feel the same as 4★ & 5★).

•Monthly salary → Ratio

- Ordered, equal differences, and true zero (0 salary = no income).
- Ratios make sense (a \$2000 salary is twice a \$1000 salary).

•Time of day (2:00 pm, 3:00 pm, 4:00 pm) → Interval

- Ordered and equal differences.
- But "0:00" (midnight) is not the absence of time.

•Marital status (single, married, divorced, widowed) → Nominal

- Just categories with no order.

•Temperature in Celsius → Interval

- Ordered, equal differences.
- No true zero ($0^{\circ}\text{C} \neq$ no temperature)

Before we move on...

Again remember

- Time
- What kind of variable is it ???



Before we move... remember

- Time
- Quantitative
- Continuous
- Interval
- Ratio ???



Before we move... remember

- Time
- Quantitative
- Continuous
- Interval
- Ratio – Yes if Duration.
- E.g. 10 min is 2x of 5 min



Frequency

- **Definition:**
- Frequency is the number of times a particular value or category occurs in a dataset.

Fruit	Frequency
Apple	5
Mango	8
Banana	4
Orange	3



Types of Frequency

- **Absolute Frequency**

- The actual number of times a value or category occurs.
- Example: If 8 students like mango, the absolute frequency of mango = **8**.

- **Relative Frequency**

- The proportion (or percentage) of the total that a category represents.
- Formula:

$$\text{Relative frequency} = \frac{\text{frequency of the event occurring}}{\text{total number of trials of the experiment}}$$

- Example: If 8 out of 20 students like mango → Relative frequency = $8/20 = \mathbf{0.40 (40\%)}$.

Types of Frequency (Contd.)

- **Cumulative Frequency**

- The running total of frequencies as you move through the categories in order.
- Example: If frequencies are Apple = 5, Mango = 8, Banana = 4 →
 - Cumulative frequency after Apple = 5
 - After Mango = $5 + 8 = 13$
 - After Banana = $13 + 4 = 17$

- **Cumulative Relative Frequency**

- The running total of relative frequencies.
- Example: If relative frequencies are Apple = 25%, Mango = 40%, Banana = 20%, Orange = 15% →
 - Cumulative relative after Apple = 25%
 - After Mango = $25\% + 40\% = 65\%$
 - After Banana = $65\% + 20\% = 85\%$

Practice

- Question 1: Favorite Colors of Students

Color	Frequency
Red	6
Blue	4
Green	5
Yellow	3
Black	2

Solution

Color	Absolute Frequency	Relative Frequency (%)	Cumulative Frequency	Cumulative Relative (%)
Red	6	$6/20 = 0.30 \rightarrow 30\%$	6	30%
Blue	4	$4/20 = 0.20 \rightarrow 20\%$	$6+4 = 10$	$30\%+20\% = 50\%$
Green	5	$5/20 = 0.25 \rightarrow 25\%$	$10+5 = 15$	$50\%+25\% = 75\%$
Yellow	3	$3/20 = 0.15 \rightarrow 15\%$	$15+3 = 18$	$75\%+15\% = 90\%$
Black	2	$2/20 = 0.10 \rightarrow 10\%$	$18+2 = 20$	$90\%+10\% = 100\%$



Practice

Question 1: Drinks Sold
in a Shop (One Day)

Solution



Drink	Frequency
Tea	12
Coffee	8
Juice	10
Soda	5
Total	35

Drink	Absolute Frequency	Relative Frequency (%)	Cumulative Frequency	Cumulative Relative (%)
Tea	12	$12/35 \approx 0.343 \rightarrow 34.3\%$	12	34.3%
Coffee	8	$8/35 \approx 0.229 \rightarrow 22.9\%$	$12+8 = 20$	$34.3\%+22.9\% = 57.2\%$
Juice	10	$10/35 \approx 0.286 \rightarrow 28.6\%$	$20+10 = 30$	$57.2\%+28.6\% = 85.8\%$
Soda	5	$5/35 \approx 0.143 \rightarrow 14.3\%$	$30+5 = 35$	$85.8\%+14.3\% = 100\%$

Use Case (Practice Problem 1)



- **Absolute Frequency (Counts)**
 - *Use:* The teacher wants to know exactly how many students like each color.
 - *Example:* "6 students like Red."
- **Relative Frequency (Percentage)**
 - *Use:* To compare which colors are most/least popular.
 - *Example:* "30% like Red, only 10% like Black."
- **Cumulative Frequency (Running Total)**
 - *Use:* To find how many students like colors up to a certain point in the list.
 - *Example:* If we count Red + Blue = 10 → "10 students like either Red or Blue."
- **Cumulative Relative Frequency (Running %)**
 - *Use:* To see what portion of students we've covered after listing some colors.
 - *Example:* After Red and Blue = 50% → "Half the class likes either Red or Blue."

Use Case (Practice Problem 2)



- **Absolute Frequency (Counts)**

- *Use:* The shopkeeper wants the exact number of drinks sold.

- *Example:* “12 cups of Tea sold.”

- **Relative Frequency (Percentage)**

- *Use:* To know customer preference in proportion.

- *Example:* “34% of sales were Tea, 14% were Soda.”

- **Cumulative Frequency (Running Total)**

- *Use:* To find how many drinks together make up most sales.

- *Example:* Tea (12) + Coffee (8) = 20 → “20 drinks were Tea or Coffee.”

- **Cumulative Relative Frequency (Running %)**

- *Use:* To measure contribution towards total sales.

- *Example:* Tea (34.3%) + Coffee (22.9%) $\approx$ 57.2% →

- “Tea and Coffee together make up more than half the sales.”

From the book

A **frequency distribution** is the organization of raw data in table form, using classes and frequencies.

Categorical Frequency Distributions

The **categorical frequency distribution** is used for data that can be placed in specific categories, such as nominal- or ordinal-level data. For example, data such as political affiliation, religious affiliation, or major field of study would use categorical frequency distributions.

EXAMPLE 2-1 Distribution of Blood Types

Twenty-five army inductees were given a blood test to determine their blood type. The data set is

A	B	B	AB	O
O	O	B	AB	B
B	B	O	A	O
A	O	O	O	AB
AB	A	O	B	A

SOLUTION

Construct a frequency distribution for the data.

A Class	B Tally	C Frequency	D Percent
A	///	5	20
B	/// //	7	28
O	/// ///	9	36
AB	///	4	16
		Total 25	100%

For the sample, more people have type O blood than any other type.

From the book

Grouped Frequency Distributions

When the range of the data is large, the data must be grouped into classes that are more than one unit in width, in what is called a **grouped frequency distribution**. For example,

EXAMPLE 2-2 Record High Temperatures

These data represent the record high temperatures in degrees Fahrenheit (°F) for each of the 50 states. Construct a grouped frequency distribution for the data, using 7 classes.

112	100	127	120	134	118	105	110	109	112
110	118	117	116	118	122	114	114	105	109
107	112	114	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	113	120	117	105	110	118	112	114	114

Source: *The World Almanac and Book of Facts*.

SOLUTION

The procedure for constructing a grouped frequency distribution for numerical data follows.

Determine the classes.

Find the highest value and lowest value: $H = 134$ and $L = 100$.

Find the range: $R = \text{highest value} - \text{lowest value} = H - L$, so

$$R = 134 - 100 = 34$$

Select the number of classes desired (usually between 5 and 20). In this case, 7 is arbitrarily chosen.

Find the class width by dividing the range by the number of classes.

$$\text{Width} = \frac{R}{\text{number of classes}} = \frac{34}{7} = 4.9$$

Round the answer up to the nearest whole number if there is a remainder: $4.9 \approx 5$.



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Thank You

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